

APIWAT NONUT

Industrial Software Engineer | Smart Factory Developer | IoT Engineer

Suratthani, Thailand | apiwat.apply@gmail.com | 092-5853800 | Line: 0925853800 | GitHub: github.com/apiwatapply-svg

PROFESSIONAL SUMMARY

Developer Engineer with 5+ years of experience building software for factories, machine-side applications, smart factory dashboards, IoT data flows, robotics automation, and AI inspection prototypes. Hands-on experience connecting machines, operators, maintenance teams, production supervisors, and management reports.

Main work: requirement analysis, database design, backend APIs, realtime dashboards, Git workflow, CI/CD checks, PM2 deployment, on-premise support, and handover notes.

CORE SKILLS

- Factory Systems: Machine monitoring, OEE, output/target, downtime analyze, maintenance system
- Frontend: React.js, Next.js, HTML/CSS, Tailwind CSS, Bootstrap
- Backend and Data: Node.js, Express.js, Python, FastAPI, SQL Server, PostgreSQL, Supabase, Prisma
- Realtime and IoT: MQTT, InfluxDB, Socket.IO, Arduino, PLC link data, MC Protocol Modbus protocol
- DevOps: Git, GitHub Actions, CI/CD, PM2, Linux server, deployment scripts, Docker basics
- AI and Automation: OpenCV, YOLO, PyTorch, Basler camera setup, ROS, ROSBridge, AGV control, LLM

PROFESSIONAL EXPERIENCE

Developer Engineer | NMB-Minebea Thai Ltd. | Thailand | Aug 2023 - Present

- Develop smart factory software and machine-side applications for production monitoring, automation, maintenance, and paperless factory work.
- Built MMS screens and reports for machine status, output, target, OEE, NG, alarms, and daily production across 207 machines, 27 machine types, and 4 factory zones.
- Designed machine data flow using MQTT, InfluxDB, SQL Server, Node.js APIs, Socket.IO, and Next.js dashboards.
- Prepared Git workflow, CI checks, deployment notes, PM2 process control, and release steps so factory applications could be deployed and supported on premise.
- Worked with production, maintenance, and factory users to turn shop-floor problems into usable software features.
- Built ROS-based automation and AI inspection prototypes using camera, lighting, image processing, YOLO classification, and SQL records.

IoT Engineer | Kubota Research & Development Asia | Thailand | Sep 2021 - Aug 2023

- Applied IoT technologies to smart agricultural machinery, drone-related systems, and tractor auto-steering concepts.
- Conducted feasibility studies, product requirement analysis, field testing, and performance validation for new product concepts.
- Wrote field-test reports and technical recommendations for product improvement.

Facilitator | Phoenix Pulp and Paper Plc Ltd. | Thailand | May 2019 - Dec 2019

- Facilitated technical and soft-skill training for employees in a factory environment.
- Supported shop-floor problem solving and built early experience in cross-functional factory communication.

PROFESSIONAL PROJECTS

Smart Factory Operations Platform | Jan 2026 - Present

Tech: Next.js, React, Node.js, Express.js, Prisma, SQL Server, Socket.IO, Arduino

- Built a factory operations platform for maintenance, preventive maintenance planning, Realtime MMS, OEE, tooling, job requests, and reports for 100 machines, 14 type, 4 area.
- Put machine master data, PM plans, maintenance records, tooling data, and report exports in one place so the maintenance team follow work more easily.
- Added live status updates and PM notification jobs so pending maintenance work was less likely to be missed.
- Benefit: production, maintenance, QC users can check OEE, PM work, job requests, tooling records, and reports in the same system instead of following separate files.

Smart Factory MMS Dashboard | Jan 2025 - Jul 2025 | GitHub: https://github.com/apiwatapply-svg/MMS_project

Tech: Next.js, React, Node.js, Express, SQL Server, Prisma, MQTT, InfluxDB, Socket.IO, PM2

- Built an on-premises machine monitoring and OEE dashboard for 207 machines, 27 machine types, and 4 factory zones.
- Created views for output, target, OEE, availability, cycle time, machine status, alarms, daily reports, and NG history.

- Reduced repeated paper and spreadsheet checking by giving production users one screen for live machine monitoring and past records.
- Helped supervisors and engineers find abnormal machine status faster by combining live machine signals with SQL Server reports.
- Benefit: production and engineering teams can see machine problems earlier and review old records without waiting for manual report preparation.

AI Defect Inspection System | Mar 2024 - Jun 2024 | GitHub: https://github.com/apiwatapply-svg/Defect_Inspection_System

Tech: Python, OpenCV, Ultralytics YOLO, PyTorch, Basler Camera, Photometric Stereo, SQL Server

- Built an inspection prototype using a Basler camera, controlled lighting, photometric stereo processing, ROI extraction, and YOLO OK/NG classification.
- Kept raw images, processed images, prediction crops, final visual results, and SQL inspection records for each inspection run.
- Benefit: inspection results are easier to review later because image files, prediction crops, and SQL records are kept together.

Factory IoT Box for Machine Data Acquisition | Dec 2023 - Apr 2024

Tech: Arduino, MQTT, InfluxDB, SQL Server, Node.js, machine signal input

- Built Arduino-based IoT Box prototypes to read signals from production machines and send machine-side data into the MMS system.
- Mapped machine signals for status, output, alarm, and running data so the dashboard could show live machine conditions.
- Tested the on-premise data flow from IoT Box hardware through MQTT, InfluxDB, SQL Server, and the web dashboard.
- Benefit: factory users collect machine data from the source instead of relying only on manual input or end-of-shift reports.
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ROS-based Automation Systems | Sep 2023 - Feb 2024 | GitHub: https://github.com/apiwatapply-svg/ROS_agv

Tech: ROS, ROSBridge, MQTT, React dashboard, LiDAR navigation, Arduino scripts

- Built AGV-related automation using ROS navigation, web dashboard control, station commands, and field-test evidence.
- Prepared source files, maps, CAD/report assets, test evidence, and maintenance notes for handover.
- Benefit: another engineer can continue testing the AGV prototype with maps, source files, and notes already prepared.

ADDITIONAL PORTFOLIO PROJECTS

- Coffee Shop POS System: built cashier checkout, live barista queue, product management, order history, receipts, dashboard metrics, and user roles.
- Booking Meeting Room: built room schedule, booking modal, admin management, sample room data, and realistic room assets.
- Restaurant POS and UNCLONED E-Commerce: hobby projects for practicing operational UI, transaction flows, admin screens, and deployment.

EDUCATION

Master of Engineering in Mechanical Engineering | Khon Kaen University | 2019 - 2021

- Thesis: System Identification and Inner-Loop Stability Control for Fixed-Wing UAVs using Meta-Heuristics. Evaluation: Excellent.

Bachelor of Engineering in Electrical Engineering | Khon Kaen University | 2015 - 2019

- Senior Project: Ping Pong Robot using LabVIEW and NI myRIO. GPA: 3.35, Second Class Honors.

CERTIFICATIONS AND LANGUAGES

- Certifications: Test Case Design Techniques; Basic STM32 Microcontroller Programmer; KOBDEMY Full-stack Developer Program; Mastering Indoor Vertical Farming for Cannabis.
- Languages: Thai native; English conversational.